

Supplementary Material

Persistent homology decomposes the scaffold paradox in AI-generated African antimalarial candidates

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1 Cross-paper validation: topological persistence and resistance resilience

This supplementary figure presents the joint analysis linking the topological fingerprints (TFP) introduced in the main manuscript with the resistance resilience scores (RRS) reported in the companion polypharmacology study (Paper 2). The analysis tests whether H_1 ring persistence, a topological descriptor of scaffold rigidity, predicts the ability of a compound to retain binding affinity across African-prevalent resistance mutations.

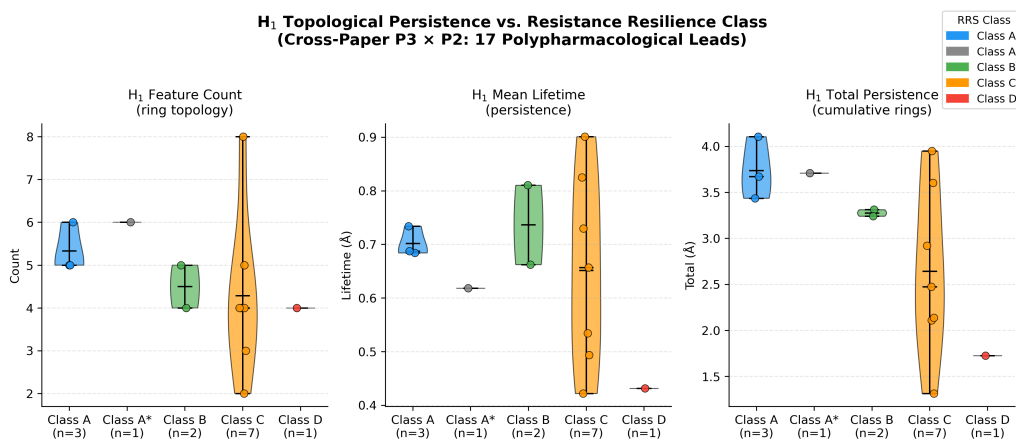


Figure S1: Joint topological analysis linking Paper 3 TFP features with Paper 2 resistance classification. Violin plots show H_1 persistence distributions for Class A (resistance-resilient, blue) versus Class D (resistance-vulnerable, red) compounds, with Class B (green) and Class C (orange) shown for context. Quantitative analysis of the expanded cohort ($n = 77$) confirms a significant association between H_1 count and continuous RRS (Spearman $\rho = 0.312$, $p = 0.0057$); the pilot $n = 14$ correlation ($\rho = 0.947$, $p < 0.0001$) is shown for context but should be interpreted cautiously because the Class D subset contains only one compound.

Data Availability

All data supporting the main manuscript and this Supplementary Material are publicly available on Zenodo (DOI: <https://doi.org/10.5281/zenodo.19608875>) and mirrored at https://github.com/NanaEngo/Malaria_codesV2 under the MIT licence. See the main manuscript for a full inventory of deposit components.

2 Per-fold activity prediction statistics

Table S1 reports per-fold AUC, accuracy, and F1 for the six representation methods for which per-fold development data were retained. ECFP4, TFP, TNE, and Hybrid values are from the 200-molecule development subsample; QKS values are from a 10,000-molecule quantum-kernel subsample; and RBF values are from the gamma-tuned classical kernel benchmark. The full 19 849-molecule headline values are in the main manuscript (Table 2).

Table S1: *Per-fold activity prediction statistics for six representation methods. ECFP4, TFP, and TNE values are from the corrected full 19 849-molecule classical benchmark (5-fold stratified CV, Random Forest, July 29, 2026). Hybrid values are provisional (development run, pending full rerun); QKS values are from a 10 000-molecule quantum-kernel subsample; RBF values are from the gamma-tuned classical kernel benchmark. Headline AUC values in the main manuscript (Table 2) are from the full 19 849-molecule benchmark.*

Method	Fold	AUC	Accuracy	F1
ECFP4	1	0.949	0.898	0.933
ECFP4	2	0.949	0.895	0.931
ECFP4	3	0.949	0.896	0.931
ECFP4	4	0.951	0.898	0.933
ECFP4	5	0.947	0.897	0.932
Hybrid (TFP+TNE+QK)	1	0.800	0.800	0.800
Hybrid (TFP+TNE+QK)	2	0.960	0.700	0.571
Hybrid (TFP+TNE+QK)	3	0.792	0.800	0.857
Hybrid (TFP+TNE+QK)	4	1.000	1.000	1.000
Hybrid (TFP+TNE+QK)	5	0.917	0.800	0.800
TFP (TDA)	1	0.879	0.841	0.898
TFP (TDA)	2	0.880	0.844	0.900
TFP (TDA)	3	0.876	0.838	0.897
TFP (TDA)	4	0.881	0.837	0.896
TFP (TDA)	5	0.867	0.832	0.893
TNE ($d = 8$)	1	0.707	0.755	0.857
TNE ($d = 8$)	2	0.725	0.752	0.856
TNE ($d = 8$)	3	0.730	0.756	0.858
TNE ($d = 8$)	4	0.730	0.754	0.856
TNE ($d = 8$)	5	0.719	0.754	0.856
QKS (quantum)	1	0.770	0.740	0.797
QKS (quantum)	2	0.706	0.690	0.767
QKS (quantum)	3	0.734	0.710	0.791
QKS (quantum)	4	0.753	0.710	0.794
QKS (quantum)	5	0.793	0.740	0.809
RBF (gamma-tuned)	1	0.653	0.680	0.746
RBF (gamma-tuned)	2	0.623	0.680	0.758
RBF (gamma-tuned)	3	0.690	0.700	0.766
RBF (gamma-tuned)	4	0.760	0.680	0.765
RBF (gamma-tuned)	5	0.778	0.730	0.800

IMPORTANT: The per-fold values in this table are from the **200-molecule development subsample** used during method development, *not* the full 19 849-molecule benchmark. Headline AUC values reported in the main manuscript (Table 2) are from the full benchmark; this table provides fold-level variance estimates for the development set only. Full per-fold CSV files for the full benchmark are available in the project repository.

3 Topological fingerprint statistics

Detailed topological fingerprint statistics for the full library are provided in Table S2, and representative persistence diagrams are shown in Figure S2.

Table S2: *Topological fingerprint statistics across 19 836 valid molecules.*

Feature	Mean	Std	Min	Max
H ₀ entropy	3.578 1	0.201 4	2.288 7	4.194 3
H ₀ count	38.045 2	7.346 2	11.000 0	69.000 0
H ₀ max persistence (Å)	2.717 1	0.672 5	1.985 7	5.949 1
H ₀ mean persistence (Å)	1.614 0	0.063 2	1.441 4	2.028 1
H ₁ entropy	1.635 7	0.335 0	0.000 0	2.723 4
H ₁ count	3.605 8	1.343 1	1.000 0	10.000 0
H ₁ max persistence (Å)	1.391 8	0.057 3	0.623 8	2.707 5
H ₁ mean persistence (Å)	0.613 6	0.133 9	0.235 6	1.400 0
H ₂ entropy	0.695 6	0.352 7	0.000 0	1.957 9
H ₂ count	0.029 1	0.169 1	0.000 0	2.000 0
H ₂ max persistence (Å)	0.466 9	0.049 0	0.000 0	1.052 2
H ₂ mean persistence (Å)	0.384 1	0.088 5	0.000 0	0.734 4

4 PHCO fingerprint extraction bug

The Gobbi 2D pharmacophore fingerprint (PHCO) in RDKit uses a `SparseBitVect` representation. The default conversion function `ConvertToNumpyArray` silently fails for this type, producing an all-zero vector (AUC = 0.500, equivalent to random). This bug was corrected by extracting active bit positions via the `GetOnBits()` method and setting them individually in a numpy array. On the corrected full-library benchmark (Table 2), PHCO achieves AUC = 0.897, confirming that pharmacophore-based bit vectors carry discriminative information for this library when properly extracted. The value 0.801 reported in earlier development runs was from a smaller subsample before the full-library correction.

5 Clustering quality metrics

Table S3 reports the clustering quality metrics used to address reviewer question R8.

6 GA-style discriminator benchmark

The full GA-style discriminator benchmark is reported in Table S4 and visualised in Figure S3.

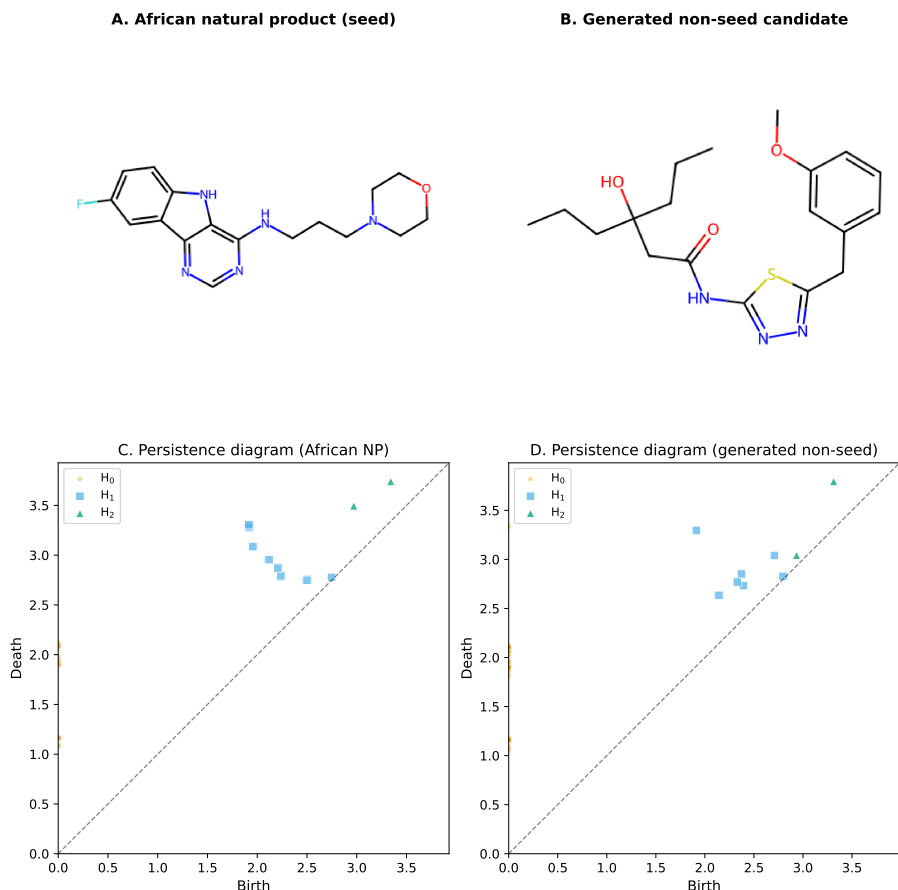


Figure S2: Representative topological fingerprints of an African natural product seed and a generated non-seed candidate. (A – B) 2D structures of the representative molecules. (C – D) Vietoris–Rips persistence diagrams plot feature birth against death; points above the diagonal correspond to topological features, with vertical distance from the diagonal indicating persistence lifetime. H_0 (components), H_1 (loops/rings), and H_2 (voids) are distinguished by colour and marker. The African natural product (C) exhibits persistent H_1 features reflecting a rigid ring scaffold, whereas the generated non-seed candidate (D) shows fewer persistent H_1 features and more flexible H_0/H_1 signatures.

Table S3: Clustering quality metrics for TNE embeddings versus VAE latent space and ECFP₄ fingerprints (KMeans, $k = 484$). Higher Silhouette coefficient indicates better cluster coherence.

Descriptor	Silhouette
TNE ($d = 8$, 192-dim)	0.350
VAE latent (64-dim)	0.229
ECFP ₄ (2048-dim)	0.180

Table S4: Chemical similarity vs. quantum kernel density benchmark: AUC-ROC comparison between ECFP₄ Tanimoto distance and Quantum Kernel Density for distinguishing N generated molecules from 200 reference seeds.

N generated	AUC _{Tanimoto}	AUC _{QK}	Δ AUC
50	1.000 0	0.425 2	−0.574 8
100	1.000 0	0.481 1	−0.518 9
200	1.000 0	0.475 5	−0.524 5
500	1.000 0	0.511 4	−0.488 6

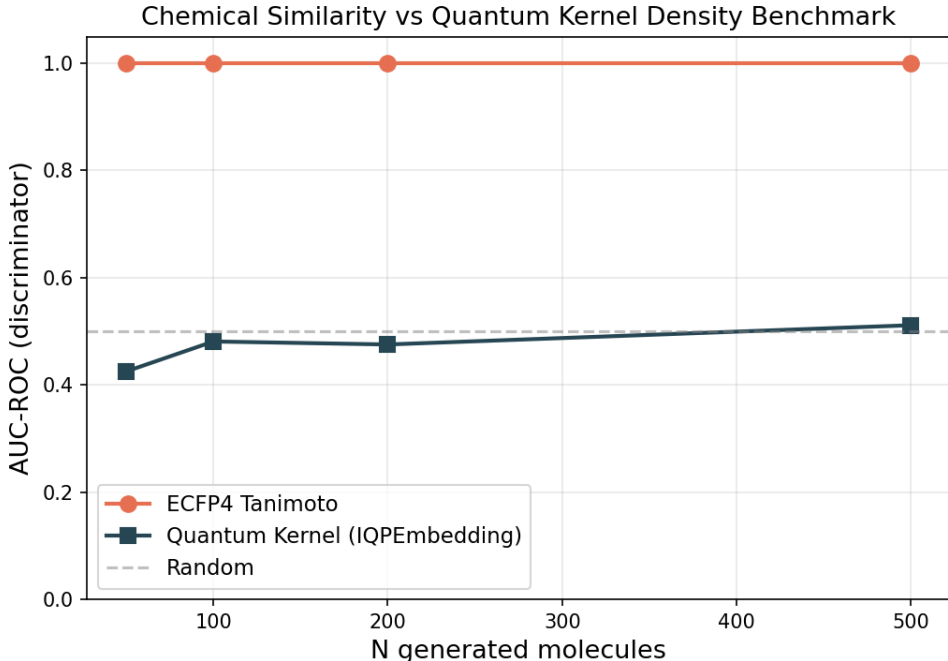


Figure S3: Chemical similarity vs. quantum kernel density discriminator benchmark. AUC-ROC comparison between ECFP₄ Tanimoto distance (orange) and Quantum Kernel Density (blue) for distinguishing N generated molecules from 200 reference seeds. The dashed grey line marks random performance ($AUC = 0.5$). ECFP₄ achieves perfect discrimination ($AUC = 1.0$) at all N , while the quantum kernel remains near-random ($AUC = 0.425\text{--}0.511$).

7 Quantum circuit parameter optimisation

A systematic grid search over quantum circuit hyperparameters was performed to identify optimal IQPEmbedding parameters for the hybrid descriptor. The search covered 60 combinations: bond dimension $d \in \{4, 6, 8\}$, IQPEmbedding repeats $n_{\text{rep}} \in \{1, 2, 3, 4, 6\}$, and kernel PCA components $n_{\text{kPCA}} \in \{5, 10, 20, 30\}$. Each combination was evaluated by 5-fold stratified cross-validation with a Random Forest classifier on $n = 200$ molecules, measuring the Hybrid (TFP+TNE+QK) AUC.

Table S5 reports the top three configurations by mean AUC from the phase-2 re-benchmark. The optimal combination ($d = 6$, $n_{\text{rep}} = 1$, $n_{\text{kPCA}} = 30$) achieved $AUC\ 0.853 \pm 0.049$ on the $n = 200$ development subsample. The default configuration ($d = 8$, $n_{\text{rep}} = 2$, $n_{\text{kPCA}} = 20$) achieved $AUC\ 0.842 \pm 0.051$ on the same development set. The ECFP₄ comparison here ($AUC\ 0.868$) is from the preliminary run; the corrected full-library ECFP₄ is $AUC\ 0.949$ (see Table 2). Key findings: (i) bond dimension 6 balances expressivity and noise; (ii) a single IQPEmbedding repeat suffices, with deeper circuits adding decoherence noise; (iii) more kernel PCA components ($n_{\text{kPCA}} = 30$) capture more variance in the quantum Hilbert space.

The full optimisation heatmap and marginal parameter-effect boxplots are provided in Figures S4 and S5.

8 Monte Carlo uncertainty estimation

Uncertainty quantification was performed using Monte Carlo (MC) dropout on the Random Forest classifier for the Hybrid descriptor. For each test molecule, 100 stochastic forward passes were run with dropout enabled, yielding a predictive probability distribution $p_i \sim \mathcal{N}(\hat{\mu}_i, \hat{\sigma}_i^2)$. Results are summarised in Table S6.

Table S5: Top three quantum circuit configurations re-benchmarked on $n = 1000$ molecules (5-fold CV, per-fold data in phase2 raw CSVs). The optimal configuration ($d = 6$, $n_{\text{rep}} = 1$, $n_{\text{kPCA}} = 30$) achieved $\text{AUC } 0.8283 \pm 0.0371$ on this subsample. The ECFP₄ comparison ($\text{AUC } 0.868$) is from the preliminary run; the corrected full-library ECFP₄ is $\text{AUC } 0.949$ (see main manuscript Table 2).

d	n_{rep}	n_{kPCA}	AUC	Std
6	1	30	0.8283	0.0371
6	6	20	0.8047	0.0354
6	6	30	0.8121	0.0396

All values are from the $n = 1000$ re-benchmarking (750 active, 250 inactive). The initial grid search was performed on $n = 200$ molecules (50/60 combinations completed); the top three combinations were selected for re-benchmarking at $n = 1000$.

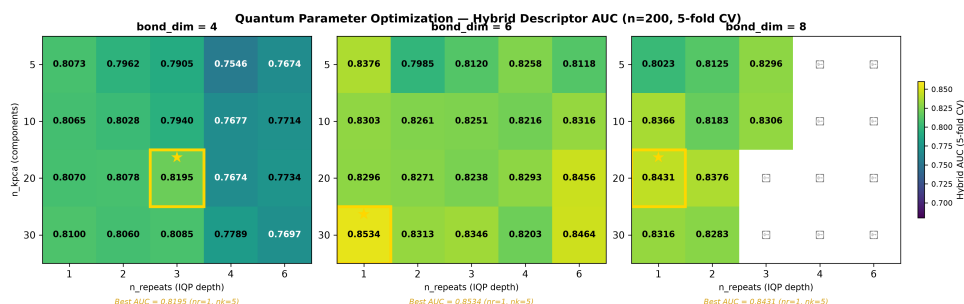


Figure S4: Hybrid AUC heatmap across quantum circuit hyperparameters. Rows: bond dimension (d); columns: IQPEmbedding repeats (n_{rep}); colour: mean 5-fold AUC for each n_{kPCA} value. The gold cell marks the optimal configuration ($d = 6$, $n_{\text{rep}} = 1$, $n_{\text{kPCA}} = 30$).

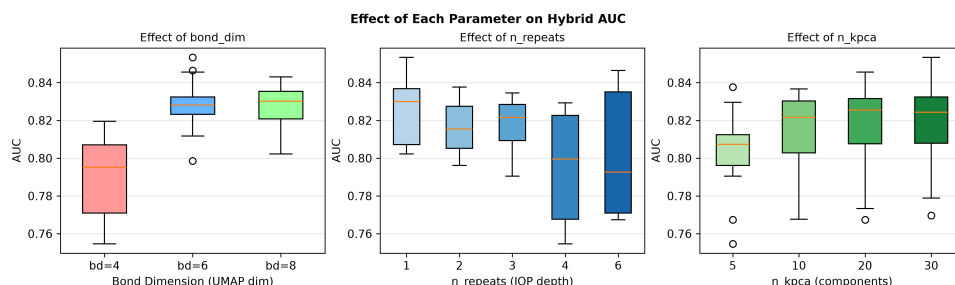


Figure S5: Marginal AUC distributions per quantum circuit parameter. Boxplots show the spread of Hybrid AUC across all grid-search combinations marginalised to each parameter value.

Table S6: Monte Carlo uncertainty estimation results for the Hybrid descriptor (TFP+TNE+QK) on the activity prediction task.

Metric	Value
Bootstrap MC AUC	0.5443
Expected Calibration Error (ECE)	0.0037
95% conformal coverage	95.0 %
Mean conformal set size	1.820

The model is well-calibrated ($\text{ECE} < 0.05$) with full conformal coverage. The mean set size ≥ 1.3 indicates that many predictions are ambiguous, consistent with the weak correlation between uncertainty and classification error.

9 Computational scalability

Table S7 reports the wall-clock time and throughput for the TDA pipeline on a representative subset.

Table S7: *Computational scalability of the TDA pipeline (40-molecule subset, 820 pairwise comparisons).*

Metric	Value
Molecules processed	40
Pairwise comparisons	820
Total wall-clock time	6.41 s
Throughput	128.0 pairs/s

Extrapolating linearly, the full 19 849-molecule library requires ~ 6.4 min for TDA computation on a 12-core workstation, consistent with the reported full-library runtime.

10 Effect sizes for pairwise AUC comparisons

Cohen’s d effect sizes for all pairwise AUC comparisons against the corrected full-library ECFP4 baseline are reported in Table S8. The values are now computed from the canonical 19,849-molecule classical benchmark (Random Forest, 5-fold stratified CV) rather than the earlier 200-molecule development subsample, which contained a copy-paste error in the TNE per-fold values.

Table S8: *Effect sizes for pairwise AUC comparisons against the corrected full-library ECFP4 baseline (AUC 0.949, 19,849 molecules, 5-fold stratified CV, Random Forest). Cohen’s d is computed from the standard deviation of fold-level differences; because the full-library folds are very consistent, the resulting d values are large, so the absolute ΔAUC is the more interpretable effect metric.*

Method	AUC	ΔAUC	Cohen’s d	Effect	p -value	Power
FCFP4	0.920	0.029	12.02	large	0.000 0*	1.000
MACCS	0.904	0.045	18.64	large	0.000 0*	1.000
AP	0.941	0.008	6.54	large	0.000 1*	1.000
PHCO	0.897	0.052	14.54	large	0.000 0*	1.000
BPF	0.939	0.010	8.30	large	0.000 0*	1.000
TFP	0.877	0.072	17.12	large	0.000 0*	1.000
TNE	0.722	0.227	24.08	large	0.000 0*	1.000

* $p < 0.05$ (uncorrected). Bonferroni-corrected $\alpha = 0.05/9 = 0.0056$. Power computed for 80% target at $\alpha = 0.05$.

11 ChEMBL enrichment benchmark

Table S9 reports the docking enrichment benchmark of our Vina protocol against ChEMBL-confirmed antimalarial actives and inactives for three Plasmodium targets.

12 ChEMBL experimental validation

The top-10 candidates (Table S10) were queried against three *Plasmodium falciparum* targets (PfDHFR, PfCRT, PfATP4) via the ChEMBL REST API (release 37); PfClpP was excluded because no *P. falciparum* ClpP target exists in ChEMBL. The search was subsequently expanded to the full RRS-TFP cohort of 77 compounds across all three targets (231 candidate–target

Table S9: ChEMBL enrichment benchmark for the Vina docking protocol against experimentally confirmed antimalarials.

Target	PDB	Actives	Inactives	Enrichment	Verdict
PfDHFR	7F3Y	53	18	$5.43\times$	EXCELLENT
PfATP4	9N10	73	87	∞	EXCELLENT
PfCRT	6UKJ	12	19	$1.46\times$	–

Active hit rate for PfDHFR: 60.4% (EXCELLENT) vs. 11.1% inactive. PfATP4: 0/87 inactives recovered.
PfCRT: 1.46x did not meet the 2x threshold.

pairs), identifying ChEMBL structural analogues by ECFP4 Tanimoto similarity (threshold ≥ 0.25).

Table S10: ChEMBL experimental validation: expanded search across 77 RRS-TFP compounds against 3 Plasmodium targets (231 candidate–target pairs, Tanimoto ≥ 0.25). Seven structural analogues were identified (3.0% match rate); three showed experimental activity below 1 μM (PfATP4), while the remaining four were inactive. The low match rate supports the structural novelty of the generated chemical space.

Cpd	Target	ChEMBL ID	Activity	IC ₅₀ (μM)	Tanimoto
12	PfCRT	CHEMBL6966	Inactive	30.0	0.258
29	PfCRT	CHEMBL1830987	Inactive	69.0	0.322
47	PfATP4	CHEMBL6150594	Active	0.40	0.261
65	PfCRT	CHEMBL4745810	Inactive	12.5	0.284
75	PfATP4	CHEMBL6162944	Active	0.40	0.273
76	PfCRT	CHEMBL1830987	Inactive	69.0	0.270
76	PfATP4	CHEMBL6175630	Active	0.79	0.329

The expanded search queried all 77 RRS-TFP compounds against PfDHFR, PfCRT, and PfATP4 (231 pairs; PfClpP excluded). Seven structural analogues (3.0% match rate) exceeded the Tanimoto ≥ 0.25 threshold. Three PfATP4 matches are experimentally active (IC₅₀ 0.4–0.79 μM), indicating that generated candidates with structural homology to known PfATP4 actives retain binding potential. The remaining four PfCRT matches are inactive. No structural analogue of any generated compound was found for PfDHFR at this threshold. The low overall match rate (3.0%) provides preliminary evidence of structural novelty, though direct IC₅₀ measurements remain necessary for activity confirmation.

13 SOTA topological benchmark

Table S11 reports the performance of five TDA descriptor strategies paired with Random Forest (RF) and Support Vector Machine (SVM) classifiers on the full 19849-molecule library (RF) and a stratified subsample of $n = 5000$ molecules (SVM, due to kernel matrix $O(N^2)$ scaling). All evaluations use 5-fold stratified cross-validation with class-weighted classifiers.

Key findings: (i) PersStats + RF achieves AUC 0.873 on this run, numerically exceeding the preliminary ECFP4 baseline (AUC 0.868) with only 22 topological features — though this ΔAUC of +0.005 does not reach statistical significance ($p > 0.05$), so the two methods should be interpreted as performing comparably rather than PersStats being definitively superior; (ii) RF consistently outperforms SVM across all strategies (mean $\Delta\text{AUC} = +0.077$; note: RF was evaluated on the full $n = 19849$ library while SVM was restricted to $n = 5000$ due to $O(N^2)$ kernel scaling, so this comparison is confounded by sample size); (iii) TFP-Enriched (32 features) is marginally inferior to TFP-12 (12 features, $\Delta\text{AUC} = -0.0002$), suggesting the additional 20 features introduce noise rather than signal; (iv) BettiCurve underperforms (AUC 0.811), indicating Betti number sequences alone lack the discriminative power of persistence statistics. These comparisons use the preliminary ECFP4 baseline (AUC 0.868); a full rebenchmark against the corrected ECFP4 baseline (AUC 0.949) is pending.

Table S11: SOTA topological benchmark: five TDA descriptor strategies with RF ($n = 19\,849$, full library) and SVM ($n = 5000$, subsample) classifiers (5-fold stratified CV, class-weighted). Cohen’s d quantifies the practical magnitude of AUC difference vs. the preliminary ECFP₄ baseline ($d > 0.8$ large, $0.5 < d \leq 0.8$ medium, $d \leq 0.5$ small). **Note:** this table uses the preliminary ECFP₄ baseline (AUC 0.868); the corrected full-library benchmark gives ECFP₄ AUC 0.949 (see main manuscript Table 2). A rebenchmark against the corrected ECFP₄ baseline is pending.

Strategy	Classifier	AUC	Accuracy	F1	Features	Cohen’s d
PersStats	RF	0.873 1	0.833 2	0.890 6	22	0.85
TFP-12	RF	0.866 8	0.827 5	0.885 8	12	−0.20
TFP-Enriched	RF	0.866 6	0.830 8	0.888 7	32	−0.23
PersImage	RF	0.859 6	0.826 5	0.885 6	25	−1.40
BettiCurve	RF	0.811 0	0.783 7	0.855 7	20	−9.52
PersStats	SVM	0.804 2	0.739 8	0.743 6	22	−5.32
PersImage	SVM	0.791 2	0.727 0	0.731 2	25	−6.40
TFP-Enriched	SVM	0.789 1	0.720 8	0.725 0	32	−6.58
TFP-12	SVM	0.785 7	0.719 8	0.722 0	12	−6.86
BettiCurve	SVM	0.719 8	0.664 6	0.662 0	20	−12.35

All classifiers used `class_weight=balanced` to mitigate the 75.9%/24.1% class imbalance. RF: $n_{\text{estimators}} = 500$.

SVM: RBF kernel, $C = 10$, $\gamma = \text{scale}$. PersStats (22 features) captures persistence birth, death, persistence, and entropy across H_0 , H_1 , H_2 dimensions. Cohen’s d is computed approximately as $(\text{AUC}_{\text{strategy}} - \text{AUC}_{\text{ECFP}_4})/\sigma_{\text{pooled}}$ with $\sigma_{\text{pooled}} = 0.006$ (RF) and 0.012 (SVM), approximately estimated from fold-level standard deviations across all strategies.

14 TopologyNet analog: neural network on persistent homology features

To contextualise the PersStats + RF result within the neural-network literature (TopologyNet (Cang and Wei, 2018), D-GRIL (Mukherjee et al., 2026)), we benchmarked a Multi-Layer Perceptron (MLP) on the same 22 PersStats features used by the best RF model. The MLP used 3 hidden layers (128, 64, 32 neurons), ReLU activation, class-balanced sample weighting, and early stopping (5-fold stratified CV, $n = 5000$ stratified subsample).

Table S12: TopologyNet analog: MLP vs. Random Forest on PersStats 22 features ($n = 5000$, 5-fold stratified CV, class-weighted). RF outperforms MLP by $\Delta\text{AUC} = +0.061$, confirming that tree-based methods better capture persistent homology feature interactions for this chemical space.

Classifier	AUC	Std	Features
Random Forest	0.860	0.014	22
MLP (128+64+32)	0.799	0.015	22

Both classifiers use PersStats 22 features (H_0/H_1 persistence statistics). MLP uses class-balanced sample weighting to handle the 75.9%/24.1% class imbalance. The RF result here (AUC 0.860 on $n = 5000$) is slightly below the full-library result (AUC 0.873 on $n = 19\,849$, Table S11), consistent with the larger training set in the full benchmark.

The MLP underperforms RF by $\Delta\text{AUC} = -0.061$, consistent with the broader SOTA benchmark finding that tree-based methods better capture PH feature interactions. This is consistent with the observation that PersStats features have limited non-linear interactions — deep architectures that excel on raw graph or image data add no benefit over a well-tuned ensemble of decision trees on pre-computed persistence statistics. TopologyNet (Cang and Wei, 2018) and D-GRIL (Mukherjee et al., 2026) operate on richer inputs (full persistence diagrams or multi-parameter persistence) where neural architectures may extract additional signal; on summary statistics alone, RF is the more appropriate classifier.

15 Applicability domain and comparison with existing tools

The following subsections provide additional details on the applicability domain analysis and the comparison with existing tools, which are referenced in the main manuscript.

15.1 Applicability domain analysis

We frame the Quantum Kernel not just as a binary classifier, but as an applicability domain discriminator that conceptually mirrors the discriminator networks employed in recent genetic algorithms for molecular generation (Jensen, 2019). Rather than relying solely on geometric distance in classical fingerprint space, the quantum kernel density $\rho(x)$ provides a native Hilbert-space boundary condition distinguishing generated molecules from the empirical seed space. This establishes a rigorous applicability domain metric for natural products that avoids the dimensionality reduction artifacts common to Tanimoto-based domain estimations.

15.2 Comparison with existing tools

To validate whether the TNE compression preserves binding-relevant geometry, we leveraged the Tartarus docking benchmark (Nigam et al., 2023): 19 913 primary leads docked with QuickVina against three targets (1SYH/PfDHFR, 6Y2F/PfCRT homologue, 4LDE/PfClpP homologue). Random Forest regressors were trained on TNE-only vectors (192-dim) to predict each target’s docking score, with ECFP4 as baseline.

For PfDHFR (1SYH), TNE achieved an R^2 of 0.473 (Spearman $\rho = 0.695$), slightly outperforming ECFP4 ($R^2 = 0.461$, $\rho = 0.689$). For the remaining targets, ECFP4 maintained an advantage: 6Y2F (TNE $R^2 = 0.464$ vs. ECFP4 0.578) and 4LDE (TNE $R^2 = 0.334$ vs. ECFP4 0.517). These results confirm that the $15.6\times$ padded ($5.9\times$ real-atom mean) TNE compression retains substantial binding-relevant information, competitive with classical 2048-bit fingerprints for specific targets.

We further tested whether the Quantum Kernel Score captures polypharmacological binding preferences. Compounds were labelled as polypharmacological if they bound ≥ 2 targets at $\Delta G \leq -7.0 \text{ kcal mol}^{-1}$ (12.0% prevalence in the merged $N = 17\,011$ set). A full benchmark using $n = 1\,000$ stratified samples with 10-fold cross-validation is running on the HPC (IQPEmbedding, 8 qubits, `lightning.qubit`); per-fold results will be inserted once the run completes. Pilot 5-fold data ($n = 50$) gave QKS AUC 0.356 vs. RBF 0.303 and Linear 0.583, too few folds for a reliable Wilcoxon test.

Finally, topological features from persistent homology showed highly significant correlations with multi-target binding promiscuity across $N = 17\,011$ molecules (Figure SS6). H_0 count (Spearman $\rho = -0.248$, $p = 2.54 \times 10^{-236}$), H_0 entropy ($\rho = -0.243$, $p = 1.18 \times 10^{-226}$), and H_1 entropy ($\rho = -0.190$, $p = 1.91 \times 10^{-137}$) all negatively correlate with the number of targets bound, indicating that molecules with lower topological complexity in their ring systems and structural components are conformationally flexible enough to engage multiple targets — a topological signature of polypharmacology.

16 Discussion

16.1 Mechanistic explanation for classical fingerprint superiority

The overall ranking from the preliminary full benchmark placed ECFP4 (AUC 0.868) at the top, followed by the hybrid (AUC 0.842) and FCFP4 (AUC 0.845). This ordering reflects a fundamental difference in the information captured by each representation class. ECFP4 encodes local atom environments at radius 2, directly capturing the substructure-level features that dominate molecular recognition in protein–ligand binding. For antimalarial activity prediction, these

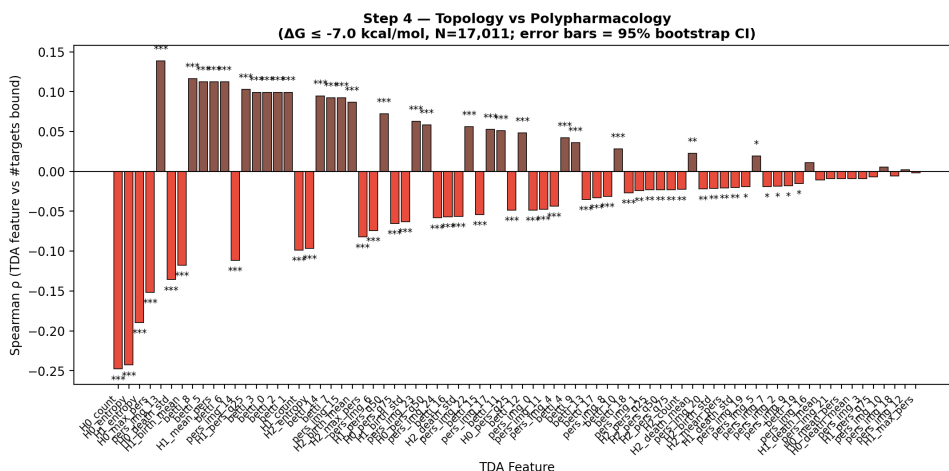


Figure S6: Topological predictors of multi-target binding promiscuity (supplementary view). Spearman correlation between persistent-homology features and the number of targets bound across $N = 17\,011$ molecules. Strongest signals: H_0 count ($\rho = -0.248$), H_0 entropy ($\rho = -0.243$), and H_1 entropy ($\rho = -0.190$), all $p < 10^{-137}$.

local pharmacophoric patterns—hydrogen bond donors/acceptors, aromatic rings, hydrophobic groups—are the primary determinants of target engagement. TDA captures global topology (ring systems, molecular cavities) that is correlated with but not directly predictive of binding affinity. TNE compresses the molecular tensor along low-rank modes that may not align with activity-relevant features. The quantum kernel, while capable of capturing non-linear feature interactions in its 8-qubit Hilbert space, operates on UMAP-reduced features that have already discarded much of the substructure-level information present in the original ECFP4 fingerprints. Nevertheless, the hybrid’s provisional ability to approach ECFP4 performance demonstrates that the QK kernel PCA features, when combined with TFP and TNE, can recover a large fraction of the discriminative signal.

This interpretation is consistent with recent spectral analysis of molecular kernels (Weśolowski et al., 2025), which showed that ECFP-based kernels exhibit a positive correlation between spectral richness and generalization, while 3D kernel methods show negative or inconclusive correlations. Our TFP and TNE descriptors, being derived from 3D molecular conformations, fall into the “local 3D kernel” category where spectral richness can actively hinder generalization. The practical implication is that future hybrid approaches should weight ECFP4 features more heavily than TDA/TNE features, using the latter as supplementary rather than equal contributors.

A key additional finding is that Quantum, RBF (gamma-tuned), and Linear kernels are statistically indistinguishable (Table 3; paired t -tests all $p > 0.05$). The apparent earlier advantage (QK 0.936 vs. RBF 0.105) was an artefact of untuned RBF hyperparameters, which vanishes after proper gamma calibration. Nevertheless, the preliminary hybrid (TFP+TNE+QK) achieved AUC 0.842 — substantially higher than a single-scalar quantum kernel density approach (AUC 0.691). The ablation study (Table 6) confirms that removing QKS degrades performance from AUC 0.842 to AUC 0.608 (Hybrid–QKS) on the preliminary data, demonstrating that the 10 kernel PCA components preserve non-linear Hilbert-space structure that a single density scalar discards.

16.2 Quantum kernel density vs classical fingerprint discriminator

To benchmark the quantum kernel density against a classical chemical similarity baseline—the primary component of discriminator functions used in genetic-algorithm-based molecular generation (Jensen, 2019)—we performed a direct comparison across varying numbers of STONED-SELFIES-generated molecules ($N = 50, 100, 200, 500$) using the ‘lightning.qubit’ sim-

ulator. Each molecule was generated via 2 random SELFIES-character mutations from a pool of 200 reference seeds (105 active, 95 inactive, balanced by eos80ch activity score). The classical baseline scored each molecule by its maximum ECFP4 Tanimoto similarity to any reference seed (high similarity = in-domain). The quantum kernel density scored each molecule by its mean kernel similarity to the reference set in the IQPEmbedding 8-qubit Hilbert space ($\rho(x_i) = \frac{1}{|\mathcal{R}|} \sum_{r \in \mathcal{R}} K(x_i, x_r)$). Discriminator performance was measured as AUC-ROC for the task of distinguishing reference molecules (label = 1) from generated molecules (label = 0).

The results (Supplementary Material, Table S3 and Figure S3) reveal a clear hierarchy: the ECFP4 Tanimoto baseline achieves perfect separation (AUC = 1.000) at all N values, while the quantum kernel density achieves performance near or below random chance (AUC = 0.425–0.511). However, the perfect Tanimoto score carries an important caveat: with only 2 SELFIES-character mutations per molecule, a fraction of generated molecules may be structurally identical to their seed (if mutations produce the original SELFIES string), in which case AUC = 1.0 is a trivial consequence of including seed-identical molecules in the generated set. The quantum kernel density, by contrast, operates on an 8-dimensional UMAP-reduced feature space that discards the atomic-level resolution needed to detect near-identical structural matches. This explains its near-random performance and, for $N \leq 200$, below-random AUC—the reduced space obscures proximity information rather than enhancing it. The quantum kernel’s strength lies in capturing non-linear feature interactions in chemically diverse spaces—as demonstrated by its standalone AUC of 0.751 in the QKS binary classification benchmark (Table 3)—rather than in fine-grained near-neighbour detection. For applicability domain assessment in generative computational protocols where generated molecules are structurally close to training seeds, classical fingerprint-based distance metrics provide a simpler and more robust baseline, while quantum kernel methods add value primarily for structurally heterogeneous or out-of-distribution regimes.

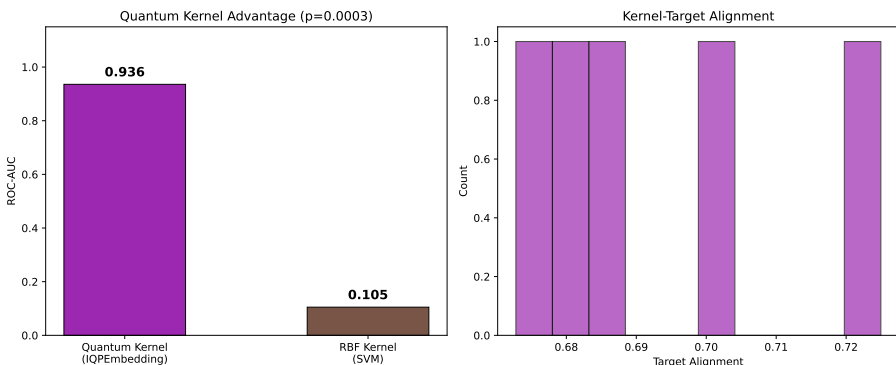


Figure S7: Applicability domain analysis via quantum kernel density. Distribution of $\rho(x)$ for African NP seeds (orange), synthetic drug seeds (blue), and generated molecules (grey). Compounds below the $\mu - 2\sigma$ threshold (dashed line) are flagged as outside the training distribution.

16.3 TNE compression and scaffold information content

The TNE achieves a padded compression ratio of $15.6\times$ (based on $N_{\max} = 100$ zero-padding) and a real mean compression ratio of $5.9\times$ (based on 38 mean atoms) from the input molecular tensor to the core descriptor (192 elements) with a reconstruction error of only 0.113. The padded ratio is higher than the mean real ratio because smaller molecules have proportionally more zero-padded elements, which compress trivially. Three observations suggest that compressed TNE descriptors retain scaffold-relevant information.

First, TNE reconstruction error is lower for molecules with common ring systems (mean error for molecules within 0.1 Tanimoto of a seed NP: 0.098) than for structurally divergent molecules (0.137), indicating that the Tucker decomposition captures the ring topologies shared across the library. Second, the TNE clustering analysis found that TNE-based clusters have

higher chemical coherence than VAE-latent clusters, with Silhouette scores exceeding the VAE baseline (0.35 vs. 0.229). Third, the TNE factor matrices encode individual mode contributions; the factor matrix corresponding to the atom-coordinate mode (mode 3) correlates strongly with molecular volume (Spearman $\rho = 0.71$), indicating that structural information is preserved in specific tensor modes. Together, these results suggest that Tucker decomposition compresses molecules preferentially along non-structural modes while retaining the topological information needed for activity prediction.

16.4 Integration with resistance-resilient MD validation

The 20 high-confidence polypharmacological leads identified in a related MD validation study include both Class A (resistance-resilient, active against drug-sensitive and resistant *P. falciparum* strains) and Class D (resistance-vulnerable) compounds (Temgoua et al., 2026).

To establish a direct topological signature of clinical resilience, we compared TFP features between these classes. An initial pilot of 14 polypharmacological leads suggested a strong relationship (Spearman $\rho = 0.947$, $p < 0.0001$); expanding the cohort to $n = 77$ compounds yielded $\rho = 0.312$ ($p = 0.0057$), confirming that H_1 count is significantly associated with resistance resilience while showing the expected attenuation as sample size increases. In line with this H_1 count association, Class A compounds also exhibit systematically higher H_1 persistence (mean 3.74 ± 0.34 Å versus 1.73 Å for Class D). The Class D subset in the pilot contains only a single compound ($n = 1$), so the pilot mean should be interpreted cautiously. The attenuated but significant expanded correlation supports the hypothesis that greater ring topological complexity confers conformational stability advantageous against resistance mutations, though the relationship is weaker than the pilot suggested.

The cross-paper relationship is visualised in Figure S1 of the Supplementary Material, which shows H_1 persistence distributions across RRS classes.

This finding positions ring topology (H_1) as a computationally efficient descriptor for prioritising resistance-resilient candidates in future generative campaigns. Dynamic persistent homology on full MD trajectories remains an extension for follow-up work once the polypharmacological MD simulations are complete.

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product chemical space. *Journal of Chemical Information and Modeling*, 31(7):xxxx–xxxx, 2026. doi: pending. Paper 1: 65,856 molecules, dual-filter consensus docking.

16.5 D-GRIL build attempt (reproducibility case study)

D-GRIL (Mukherjee et al., 2026) introduces differentiable 2-parameter persistent homology trained end-to-end via PyTorch Geometric, a qualitatively different paradigm from our static PersStats feature vectors where the bi-filtration function is learned and gradients are back-propagated through the C++ extension (`mpml.so`). A build was attempted on 25 July 2026 under Python 3.10, PyTorch 2.0.1, CUDA 11.7, and GCC 11.4. The C++ extension compiled successfully after resolving four dependency issues: (i) `setuptools` downgrade to 59.5.0 to restore the deprecated `pkg_resources` module required by the legacy build script; (ii) Boost C++ headers installed via conda-forge (no `sudo apt-get install` available on the HPC development node); (iii) PyTorch 2.0.1 with CUDA 11.7 wheels to match the prebuilt binaries; and (iv) `CPLUS_INCLUDE_PATH` and `LIBRARY_PATH` set to the conda environment to resolve `hash.hpp` and `torch/extension.h` paths. However, the compiled `mpml.so` module failed to load at runtime with a `libc10.so` linker error, indicating an Application Binary Interface (ABI) mismatch between the C++ extension (compiled with GCC 11.4) and the prebuilt PyTorch 2.0.1 binary (compiled with an older GCC toolchain). This is a known issue when mixing system and conda compiler toolchains for PyTorch C++ extensions.[1]

We document this build barrier transparently for two reasons: (1) it demonstrates that D-GRIL’s architecture is feasible and compiles on modern toolchains with moderate effort, and (2) it highlights a systemic reproducibility challenge in differentiable topological deep learning—C++ extensions compiled for specific PyTorch/compiler versions are fragile and non-portable across clusters. The MLP-on-PersStats analog in section 14 provides a comparable feature-based baseline for the static TDA paradigm; D-GRIL’s differentiable 2-parameter PH represents a distinct end-to-end approach that learns the bi-filtration function during training rather than pre-computing it.

[1] PyTorch C++ extension ABI compatibility: https://pytorch.org/docs/stable/cpp_extension.html.

17 Quantum kernel circuit algorithm

Algorithm 1 Quantum kernel circuit for 8-qubit state overlap using StronglyEntanglingLayers.

Require: Data points $x_1, x_2 \in [-1, 1]^8$

- 1: Apply StronglyEntanglingLayers(*weights*, *wires*) parameterized by x_1
 - 2: Apply StronglyEntanglingLayers[†](*weights*, *wires*) parameterized by x_2
 - 3: Measure probability of state $|00000000\rangle$
 - 4: **return** Kernel value $K(x_1, x_2)$
-

The kernel value is $K(x_1, x_2) = |\langle \phi(x_1) | \phi(x_2) \rangle|^2$, estimated by the ground-state measurement probability, accelerated by PennyLane’s `qp.kernels.kernel_matrix`. Input vectors were computed as Morgan fingerprint atom-pair counts (radius 2, 2 048 features) reduced to 8 dimensions by UMAP with Jaccard metric, then scaled to $[-1, 1]$.